

In an operating system three CPU-intensive processes are ready for execution which require 10 ns, 20 ns and 30 ns and arrive at times 0 ns, 2 ns, and 6 ns, respectively. Write a Program to calculate:

The total number of context switches needed if the operating system implements a Shortest Job First (Preemptive) scheduling algorithm.

The average waiting time of the processes before getting the CPU.

CODE:

```
1 #include <stdio.h>
2
3 int main() {
4
5     int n = 3;
6     int at[] = {0, 2, 6};
7     int bt[] = {10, 20, 30};
8     int ct[] = {10, 30, 60};    // Completion times (from SRTF)
9
10    int wt[3], total_wt = 0;
11
12    for(int i = 0; i < n; i++) {
13        wt[i] = ct[i] - at[i] - bt[i];
14        total_wt += wt[i];
15    }
16
17    printf("Process\tArrival\tBurst\tWaiting\n");
18
19    for(int i = 0; i < n; i++) {
20        printf("P%d\t%d\t%d\t%d\n", i+1, at[i], bt[i], wt[i]);
21    }
22
23    printf("\nAverage Waiting Time = %.2f ns\n", (float)total_wt / n);
24    printf("Total Context Switches = 2\n");
25
26    return 0;
27 }
```

OUTPUT:

```
Process Arrival Burst   Waiting
P1  0    10    0
P2  2    20    8
P3  6    30   24
```

```
Average Waiting Time = 10.67 ns
Total Context Switches = 2
```

```
=== Code Execution Successful ===
```